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27 August 1979
SUPERSEDING
MIL-G-45006D
19 February 1971

MILITARY SPECIFICATION

GRINDING MACHINES, CRANKSHAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers floor type, electrically powered, grinding machines, complete with equipment and accessories, to be used in grinding main and throw journals of crankshafts removed from piston type engines.

1.2 Classification. Unless otherwise specified (see 6.2), the grinding machines covered by this specification shall be of the following sizes and types:

Size 1.

Type A-Throwhead and Counterbalancing Type: 60-inch minimum capacity between centers, 18-inch minimum swing, maximum stroke not less than 7 inches.

Type B-Non Throwhead and non-counterbalancing, centerless Grinding Type: 64-inch minimum capacity between headstock chuck and tailstock centers, 13.5 inch minimum overall crankshaft diameter capacity, maximum stroke not less than 7.5 inches.

Size 2. - 74 inch minimum capacity between centers, 22 inch minimum swing, maximum stroke not less than 8.75 inches.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Materiel Readiness Command, ATTN: DRSAR-LEE-S, Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-1807 - Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).

MILITARY

MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-S-12514 - Starters, Motor; Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric
MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of
MIL-I-24092 - Insulating Varnish, Electrical, Impregnating, Solvent Containing

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA)

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments	Occupational Safety and Health Standards

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

201.02 - Tooth Proportions for Coarse-Pitch Involute Spur Gears
390.03 - Gear Classification Manual

(Application for copies should be addressed to the American Gear Manufacturers Association, One Thomas Circle, Washington, D. C. 20005.)

JOINT INDUSTRIAL COUNCIL (JIC)

EGP-1 - Electrical Standards for General Purpose Machine Tools

(Application for copies should be addressed to the Secretary of the Joint Industrial Council, care of the National Machine Tool Builders Association, 2139 Wisconsin Avenue, N. W., Washington, D. C. 20007.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B74.13 - Markings for Identifying Grinding Wheels and Other Bonded Abrasives
B7.1 - Use, Care and Protection of Abrasive Wheels, Safety Code for the

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish one complete grinding machine for first article inspection. The first article may be either a preproduction model or a production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The grinding machines shall be new and of the manufacturer's latest design that will meet the requirements specified

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herein. Sizes shall be as specified in 1.2. The grinding machines shall permit changing from line or main journal grinding to throw or rod journal grinding without removing the crankshaft from the machine. The equipment shall include all standard features, components, and accessories supplied on the manufacturer's current production of the sizes specified. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Maintainability. Design and configuration of the grinding machines, accessories, and adjustment features shall insure that maintenance of operational capability and accuracy will be reduced to the simplest form or procedures and will require a minimum of skills and common maintenance equipment. Instructions for disassembly, repair, assembly, and adjustment shall be clear, concise, and definitive in application.

3.3 Safety and health requirements. All rotating or reciprocating parts, all parts subject to high operating temperatures, and all parts which are electrically energized and are of such a nature or so located as to be a hazard to the safety of operating personnel, shall be guarded, enclosed, or insulated. All guards and covers shall be removable to provide access to the guarded part for maintenance. The grinding machines shall comply with standards promulgated under OSHA, Chapter XVII, Part 1910, with amendments to date, which are applicable to the grinding machines themselves. Additional requirements for safety and health shall be specified (see 6.2 and 6.3).

3.4.1 Parts and materials. All parts and materials used in fabrication may be virgin, recycled or reclaimed materials free of defects and imperfections that might affect safety, performance, serviceability, or appearance of the finished product. Contact between dissimilar metals, especially aluminum or magnesium in contact with steel, copper, or brass, shall be avoided wherever possible.

3.4.2 Interchangeability. All parts shall be manufactured to such standards as will permit replacement and adjustment without modification of parts or machine.

3.4.3 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No process shall be used for reclaiming any defective part.

3.4.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the orginal design. These operations shall not be employed as a repair measure for defective parts.

3.4.5 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

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3.4.6 Threads. All threaded parts used on the grinding machines and their attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.4.7 Gears. All gears and pinions shall be fabricated in the English system and shall conform to AGMA 390.03. Spur gear tooth form shall conform to AGMA 201.02.

3.4.8 Bearings. The bearings for wheelhead, headstock, and tailstock shall be lubrication sealed antifriction type, or continuously lubricated plain bearings.

3.4.9 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the connections shall be mechanically secured before soldering. Rosin base fluxes only shall be used in soldering operations. Connections to screw type terminals shall be mechanically secured with means to prevent loss of tightness.

3.5 Performance and product characteristics.

3.5.1 Capacities.

3.5.1.1 Size 1. Unless otherwise specified (see 6.2), the size 1 type A grinding machine shall be capable of grinding main and throw journals on crankshafts up to and including 60 inches in length and having a stroke up to and including 7 inches. The size 1 type B grinding machine shall be capable of grinding main and throw journals on crankshafts up to and including 64 inches in length and having a stroke up to and including 7.5 inches.

3.5.1.2 Size 2. Unless otherwise specified (see 6.2), the size 2 grinding machine shall be capable of grinding main and throw journals on crankshafts up to and including 74 inches in length and having a stroke up to and including 8.75 inches.

3.5.2 Accuracy. The grinding machines shall be capable of grinding journals in accordance with the following:

(a) Diameters ground shall be accurate in size to within plus or minus 0.0003 inch.
(b) Diameters ground shall be concentric to within 0.0003 inch total indicator reading (tir).
(c) Maximum allowable taper of ground surface shall be 0.0003 inch per foot.
(d) Final ground journal surfaces shall have a maximum surface roughness value of 16 microinches, arithmetical average (aa).

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(e) Alignment of crankshaft main journals in relation to the center axis of the crankshaft shall be within 0.0002 inch tir.

3.6 Details of components.

3.6.1 Base and bed. The base and bed shall be of cast iron or steel alloy casting, or steel plate welded fabrication at the option of the supplier. Beds shall be seasoned or stress relieved. The base and bed shall provide support and alignment for the wheelhead, headstock, and tailstock assemblies.

3.6.1.1 Bedways. All bedways shall be finished to provide an accurate, uniform bearing surface for the matching ways. All V-way and flat-way construction shall be scraped or ground. Deviation from a plane surface shall not exceed 0.001 inch over the entire length of the ways.

3.6.2 Wheelhead. The wheelhead housing shall be of sufficient proportions, weight, and strength to provide rigid support at all grinding speeds within the capability of the machine. The housing shall be constructed to accommodate the grinding wheel spindle, spindle drive, and the grinding wheel feed mechanism. The spindle shall be a heavy duty type made of alloy steel. The spindle shall be mounted by means of super-precision pre-loaded bearings or by continuously pressure lubricated plain bearings (see 3.4.8). A wheelguard of sufficient size to accommodate the grinding wheels as specified in sizes 1 and 2 machines shall be furnished. The wheel mounting assembly shall provide means for attaching wheel balancers.

3.6.2.1 Wheelhead feed. The grinding wheel cutting feed shall be manually controlled. The cutting feed control dial shall be graduated in increments of 0.001 inch to indicate removal of 0.001 inch of stock from the diameter of the journals. Rapid approach and retraction shall be provided by a mechanism with conveniently located control.

3.6.3 Headstock. The headstock housing shall be constructed to accommodate the spindle, spindle drive, faceplates, throwheads and counterbalancing weights as required for the size and type of grinder specified. The headstock spindle shall be mounted on not less than two bearings (see 3.4.8). The headstock spindle shall be designed and constructed sufficiently strong, rigid, and wear resistant to resist deflection, vibration, and wear when the grinding machine is operated at its full capacity. The spindle drive shall be stepped or varible speed type. The headstock of the size 1-type A grinder shall be designed with 360-degree rotating head and cross slides, with

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a final adjustment device for ease of crankshaft set-up. The headstock of the size 1 type B grinder shall be equipped with a three-jaw self-centering chuck with micro radial adjustment for quick crankshaft set-up.

3.6.3.1 Spindle speeds. The spindle shall have a minimum of three speeds. The low speed shall be not more than 45 revolutions per minute (rpm) and the high speeds shall be not less than 60 rpm; high speed to low speed ratio not less than 2 to 1. An intermediate speed shall be provided at midpoint between the high and low speeds. A variable speed method is acceptable provided it includes the speed range specified above.

3.6.4 Tailstock. The tailstock housing shall be constructed to accommodate the spindle, face plates, throwheads, and counterbalancing weights as required for the size and type of grinder specified. The tailstock unit shall be capable of being locked at any point along the bed or table. The size 1 type B grinder shall incorporate an automatic hydraulic tailstock lift feature which shall allow the tailstock to slide easily for length adjustments. The tailstock spindle shall be mounted on not less than two bearings (see 3.4.8). The tailstock spindle shall be designed and constructed sufficiently strong, rigid, and wear resistant to resist deflection, vibration, and wear when the grinding machine is operated at its full capacity. The spindle of the size 1, type B grinder shall have a minimum of 2 inches of travel, with provision for locking at any point of extension. The tailstock shall be at least 90 percent fully seated on the matching ways during grinding of crankshafts of specified maximum size. The tailstock of the size 1 type A grinder shall be designed 360 degree rotating head and cross slides, with a final adjustment device for ease of crankshaft set-up. The tailstock of the size 1 type B grinder shall be equipped with a three-jaw self-centering chuck with a micro radial adjustment for quick crankshaft set-up.

3.6.5 Table. When the grinding machine is provided with a traversing table, it shall provide adequate support and alignment for the headstock, tailstock, steady rests, and other components as required.

3.6.6 Centering gage. A crankshaft centering gage of the indicator or V-type for centering crankshaft journals shall be provided. The gage shall be capable of accommodating any crankshaft within the specified capacity of the grinding machine.

3.6.7 Coolant system. A complete coolant system shall be provided including power driven pump, discharge and recovery system, strainer, splash guards, piping, fittings, valves, sump or reservoir, and controls

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necessary to deliver coolant to point of contact between the grinding wheel and work surface. The flow rate of coolant shall be not less than 6 gallons per minute. The pump motor shall be as specified in 3.6.9.1. The coolant tank reservoir capacity shall be not less than 30 gallons.

3.6.8 Traversing mechanism(s). The traversing mechanism(s) for the wheelhead assembly, the traversing table, or both, depending upon machine design, shall provide for rapid traverse and slow manual feed. It (they) shall be hand operated or powered by electrical or hydraulic means with auxiliary manual control.

3.6.9 Electrical equipment. Electrical components including relays, starters, switches, wiring, and their enclosures shall be in accordance with JIC EGP-1 except as specified herein. No component part shall be used where the requirements exceed the manufacture's rating for the part.

3.6.9.1 Motors. Unless otherwise specified (see 6.2), motors shall be continuous duty, induction squirrel-cage type, with class B insulation, for operation on nominal 230/460 volts (initially wired for 230 volts), 3 phase, 60 hertz, and shall conform to CC-M-1807. The wheelhead motor shall be controlled by a starter conforming to type 1, class 2 requirements of MIL-S-12514. The headstock motor shall be provided with overload protection. Motors shall be splash-proof and mounted in accordance with JIC EGP-1. Motors shall be protected from dirt and grit and shall be accessible for lubrication and maintenance.

3.6.9.2 Motor ratings. Unless otherwise specified (see 6.2), horsepower (hp) ratings shall be not less than 5 hp for the wheelhead drive motor and not less than .5 hp for the headstock drive motor.

3.6.10 Throwhead index pins. Conveniently located index pins, or similar means, for positioning the holding the headstock and tailstock throwheads in vertical alignment shall be provided.

3.6.11 Taper adjustment. Means shall be provided to allow for proper taper adjustment and to correct for excessive taper conditions.

3.7 Standard equipment. Unless otherwise specified (see 6.2), standard equipment shall be furnished with each grinding machine as stated herein.

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3.7.1 Chucks. Two 8 inch minimum diameter, three-jaw, universal, geared scroll chucks shall be provided with each size 1 type A grinding machine. One 6 1/2-inch minimum diameter, three-jaw, universal, geared scroll headstock chuck, and one 11 inch minimum diameter, three-jaw, universal, geared scroll tailstock chuck shall be provided with each size 1, type B machine.

3.7.2 Centers. One set of three centers, hardened and ground to within 0.0002 inch tir for concentricity, shall be provided with each grinding machine.

3.7.3 Grinding wheels. Grinding wheels of the commercial type used in crankshaft grinding shall be provided for the grinding machines specified in 1.2. The grinding wheels shall be marked, handled, mounted, and used in accordance with ANSI B74.13 and ANSI B7.1, and of the sizes and quantities specified in 3.7.3.1 and 3.7.3.2.

3.7.3.1 Grinding wheel (size 1). One grinding wheel of not less than 22 inch diameter with 1 inch face, and one grinding wheel of not less than 22 inch diameter with a 1.5 inch face shall be provided for each size 1 machine.

3.7.3.2 Grinding wheel (size 2). One grinding wheel of not less than 28 inch diameter with a 1 inch face, and one grinding wheel of not less than 28 inch diameter with 1.5 inch face shall be provided for each size 2 machine.

3.7.4 Steady rests. Steady rests shall be provided to enable the workpiece to withstand being deflected from revolving on its true axis. The steady rests shall provide support for the crnakshaft journals at two points by means of individual adjustable jaws and shall be furnished as specified in 3.7.4.1 and 3.7.4.2. An integral trueing indicator, or provision for the centering gage, shall be furnished with the steady rest.

3.7.4.1 Steady rest (size 1). A minimum of two steady rests of at least 4 inch capacity shall be furnished for proper support of crankshafts for the size 1 machine.

3.7.4.2 Steady rest (size 2). A minimum of three steady rests of at least 5 inch capacity shall be furnished for proper support of crankshafts for the size 2 machine.

3.7.5 Work sizing gage. A work sizing gage designed for constant journal gaging shall be provided with each grinding machine. The gage shall consist of two adjustable caliper capacities, 1 to 3 inch and 3 to 5 inch.

3.7.6 Wheel dressing fixture. A fixture, or set of fixtures, for dressing the face, sides, and corner radii of the grinding wheel shall be provided with each grinding machine. A commercial diamond of not

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less than one carat in weight shall be mounted and rigidly supported in the fixture.

3.7.7 Throwheads. All throwheads, centers, faceplates, and driving dogs necessary to center throw journals shall be so designed and arranged so that they permit adjustment and locking alignment to within a maximum allowable error of 0.001 inch.

3.7.8 Balancing system. An integral adjustable counterbalancing system for use with the headstock, tailstock, and throwheads shall be provided and be capable of balancing any crankshaft within the rated capacity of the grinding machine.

3.8 Accessories. When specified (see 6.2), the following accessory equipment shall be furnished with each grinding machine:

(a) Grinding wheel arbor and balancing stand, or an integral wheel balancing device within the grinding machine.
(b) Portable abrasive finisher, with provisions for storing in or on the grinding machine when not in use, capable of surface finishing crankshaft journals to 3 to 7 microinch finish aa. It shall be furnished with standard equipment and three each 400 grit abrasive belts. The electric motor rating shall be not less than .25 hp.
(c) One wrench set including all necessary special tools for setting up work on the grinding machine.

3.9 Spare parts. When specified (see 6.2), the following spare parts shall be furnished with each grinding machine:

(a) One set of extra driving belts.
(b) One set of extra grinding wheels for the machine size specified (see 3.7.3.1 or 3.7.3.2).

3.10 Fungus control. When specified (see 6.2), electrical components and circuit elements shall be treated for fungus resistance in accordance with type II requirements of MIL-T-152, except that the statement of treatment is not required. Windings of the electric motors shall be given two impregnating coats of varnish conforming to the requirements of MIL-I-24092, type M, class 130 through class 220 (manufacturer's option).

3.11 Painting. Unless otherwise specified (see 6.2), painting and finishing of the grinding machine shall be in accordance with the supplier's standard practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall have been cleaned to be free of all foreign matter detrimental to painting, and at least one coat of primer and one coat of enamel or equivalent have been applied.

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3.12 Identification marking. The grinding machine shall be marked for identification in accordance with MIL-STD-130 and, when specified (see 6.2), shall include the National Stock Number.

3.13 Lubrication chart or plate. A lubrication chart or plate shall be securely attached to the grinding machine. The chart shall be contained in transparent, permanently sealed sheets with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application
Service interval
Type of lubricant
Viscosity
Federal or military specification number

3.14 Workmanship. Standards of workmanship shall be such as to assure equipment of a neat appearance, having the stability and efficient operating characteristics found in first class commercial equipment of the kind specified herein. In no event shall patching processes be used for reclaiming defective parts. All surfaces of all parts shall be cleaned and free of extraneous materials. External surfaces shall be smooth and free of sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production item noted in 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.3 and all tests under 4.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

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4.2.2 Quality conformation inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production units submitted for acceptance under the contract and shall consist of (a) through (c) below. Failure of any unit to pass any examination, test or inspection shall be cause for rejection.

(a) Product examination (see 4.3).
(b) Tests (see 4.4).
(c) Packaging inspection (see 4.5).

4.2.2.1 Inspection lot. For quality conformance inspection, a lot shall consist of all grinding machines from one manufacturer submitted for acceptance at one time.

4.2.2.2 Sampling. When applicable (see 6.2), sampling for lot acceptance shall be in accordance with MIL-STD-105, General Inspection Level II and at the Acceptable Quality Levels (AQL) specified. One grinding machine from each inspection lot shall be selected at random and tested as specified in 4.4, Tests.

4.3 Product examination. The grinding machine and its accessories shall be examined to determine compliance with the requirements of section 3 of the specification herein and the contract, using the following checklist as a guide:

EXAMINATION CHECKLIST

CHARACTERISTICS	REQUIREMENT PARAGRAPH
Design	3.2
Maintainability	3.2.1
Safety and health requirements	3.3
Construction	3.4
Parts and materials	3.4.1
Interchangeability	3.4.2
Castings and forgings	3.4.3
Welding, brazing or soldering	3.4.4
Fastening devices	3.4.5
Threads	3.4.6
Gears	3.4.7
Bearings	3.4.8
Electrical connections	3.4.9
Capacities (Size 1 or Size 2)	3.5.1
Accuracy	3.5.2
Base and bed	3.6.1
Bedways	3.6.1.1
Wheelhead	3.6.2

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EXAMINATION CHECKLIST (Cont'd)

CHARACTERISTICS	REQUIREMENT PARAGRAPH
Wheelhead feed	3.6.2.1
Headstock	3.6.3
Spindle speeds	3.6.3.1
Tailstock	3.6.4
Table	3.6.5
Centering gage	3.6.6
Coolant system	3.6.7
Traversing mechanism(s)	3.6.8
Electrical equipment	3.6.9
Motors to operate on 230/460 VAC, 3 Phase, 60 HZ, Initially wired for 230 VAC	3.6.9.1
Motor ratings	3.6.9.2
Throwhead index pins	3.6.10
Taper adjustment	3.6.11
Chucks	3.7.1
Centers	3.7.2
Grinding wheels (Size 1 or Size 2)	3.7.3
Steady rests (Size 1 or Size 2)	3.7.4
Work sizing gage	3.7.5
Wheel dressing fixture	3.7.6
Throwheads	3.7.7
Balancing system	3.7.8
Accessories	3.8
Spare parts	3.9
Fungus control	3.10
Painting	3.11
Identification marking	3.12
Lubrication chart or plate	3.13
Workmanship	3.14

4.4 Tests.

4.4.1 Operational test. The grinding machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms and other accessories shall be verified.

4.4.2 Accuracy test. The surfaces of flat and V-ways shall be tested for accuracy and alignment to determine compliance with the requirements of 3.6.1.1.

4.4.3 Headstock spindle speed test. Headstock spindle speeds shall be tested by tachometer. The low, intermediate and high speeds, or the continuously variable speed, as applicable, shall be within the limits specified in 3.6.3.1.

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4.4.4 Grinding position changeover test. The grinding position changeover from main journal grinding to throw journal grinding, as specified in 3.2, shall be accomplished during the grinding test in 4.4.5. Centering shall be checked by means of an indicator or V-type crankshaft centering gage. Centering of throw journals shall be within 0.001 inch of true center (see 3.7.7).

4.4.5 Grinding test. Unless otherwise specified (see 6.2), a used reclaimable crankshaft not less than 54 inches in length for size 1, or not less than 68 inches in length for size 2, shall be set up in the applicable size machine and the main journals and the throw journals ground. The diameter, roundness, taper and surface roughness value of journals shall be within the limits specified in 3.5.2.

4.4.5.1 Alignment test. Alignment of the crankshaft main journals ground in accordance with 4.4.5 shall be checked by setting the crankshaft in V-blocks, located on each of the end main journals. A dial indicator shall be used to check all main journals for true alignment. The tir shall be within the limit specified in 3.5.2(e).

4.5 Packaging inspection. Cleaning, preservation, packing, and marking requirements shall be inspected to determine compliance with section 5.

5. PACKAGING

5.1 Packaging (preservation, packing and marking). Level A and B shall be in accordance with the requirements of MIL-M-18058, as specified (see 6.2).

5.2 Commercial. Commercial packaging, when specified (see 6.2), shall be in accordance with the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. The grinding machines covered by this specification are floor type machines with equipment and accessories for use in grinding the main and throw journals of crankshafts of piston type engines.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Size and type required (see 1.2).
(c) First article, when required (see 3.1).
(d) Special safety and health devices, when required (see 3.3).

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(e) Capacity, size 1 or size 2 grinding machine, if different (see 3.6.9.1).
(f) Motor types and voltage characteristics, if different (see 3.6.9.1).
(g) Motor ratings, if different (see 3.6.9.2).
(h) Standard equipment, if different (see 3.7).
(i) Accessories, when required (see 3.8).
(j) Spare parts, when required (see 3.9).
(k) Fungus control, when required (see 3.10).
(l) Painting, if different (see 3.11).
(m) National Stock Number, when required (see 3.12).
(n) Sampling, when required (see 4.2.2.2).
(o) Specify level A or B (see 5.1).
(p) Specify Commercial packaging (see 5.2).

6.3 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with OSHA limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements, should be specified.

Custodians:
Army - AL
Air Force - 99

Review activity:
Army - SM

User activity:
Army - AT

Preparing activity:
Army - AL

Project No. 4910-0475

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Command
Rock Island, IL 61299

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DOCUMENT IDENTIFIER AND TITLE

MIL-G-45006E Grinding Machines, Crankshaft

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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